CHALLENGE
YOUR PRACTICES

30-Day Sustainability Action Plan

PAPER REDUCTION

SUSTAINABLE DEVELOPMENT GOALS (SDGS)

RECYCLING

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Challenge Your Practices, Detailed 30-Day Action Plan



This report presents a structured 30-day sustainability action plan aimed at promoting environmentally responsible practices within the workplace. The objective is to identify practical strategies that reduce resource consumption, minimize environmental impact, and support global sustainability goals.

The selected practices focus on key areas of environmental concern, including paper usage reduction, alignment with Sustainable Development Goals (SDGs), and the implementation of recycling systems. These practices were chosen due to their direct impact on improving resource efficiency and encouraging sustainable behavior in daily operations.

Through data analysis, simulation using MATLAB, and a step-by-step action plan, this report demonstrates how small behavioral and operational changes can contribute to long-term environmental improvement.

As part of my sustainability challenge, I selected three practices that can create a positive environmental impact in the workplace. These practices focus on reducing waste, supporting sustainability goals, and improving resource management.

Practice 1_Paper Reduction

Reduce Paper Usage by 20%



Paper consumption in workplaces remains one of the significant contributors to environmental waste and resource depletion. Reducing paper usage is an essential sustainability practice as it helps conserve natural resources, decrease waste generation, and lower operational costs associated with printing and document management.

This practice aims to achieve a 20% reduction in paper consumption over a 30-day period by encouraging the adoption of digital alternatives and promoting more efficient printing behaviors. The following action plan outlines the steps that will be implemented to achieve this goal in a structured and measurable way.

- **Action Plan**

First, I will assess the current level of paper usage in my department by collecting data on printing and paper consumption. This will help me establish a baseline for measuring progress.

After that, I will encourage employees to switch from printed documents to digital alternatives such as PDFs, shared folders, emails, and cloud-based storage systems. I will also promote double-sided printing whenever printing is necessary.

Use electronic signatures for routine approvals to reduce the need for paper.

Reduce unnecessary print jobs and replace paper-heavy processes with digital alternatives.

In addition, I will raise awareness among team members about the environmental impact of excessive paper use and encourage them to print only when absolutely needed.

This MATLAB code simulates a 30-day sustainability challenge focused on reducing resource consumption in three main categories: paper usage, electricity usage, and plastic usage. The data is generated as a set of decreasing percentage values over time to represent a 20% reduction goal.

The program begins by defining a time vector representing key checkpoints over 30 days. Then, three datasets are created to simulate usage trends for paper, electricity, and plastic. Each dataset starts at 100% and gradually decreases to 80%, reflecting the targeted reduction.

A function is defined using an anonymous function to calculate the percentage reduction between the initial and final values for each category. This allows automatic computation of the achieved improvement.

Finally, the data is visualized using a MATLAB dashboard with three subplots. Each subplot represents one resource category and shows the trend over time. The graphs are enhanced using customized line styles, markers, colors, grid lines, and titles to improve readability and presentation quality.

MATLAB CODE

```
clc;
clear;
close all;

% DATA

days = [0 7 14 21 30];

paper = [100 95 90 85 80];
electricity = [100 92 88 84 80];
plastic = [100 96 92 86 80];

calcReduction = @(x) ((x(1)-x(end))/x(1))*100;

paper_red = calcReduction(paper);
elec_red = calcReduction(electricity);
plastic_red = calcReduction(plastic);


% STYLE SETTINGS

lw = 2.5;
ms = 8;

figure('Color','w','Position',[100 100 1200 400]);

tiledlayout(1,3,'Padding','compact','TileSpacing','compact');

% PAPER

nexttile;
plot(days, paper, '-o', ...
    'LineWidth', lw, ...
    'MarkerSize', ms, ...
    'MarkerFaceColor',[0.2 0.6 1]);

grid on;
ylim([70 105]);
xlabel('Days','FontWeight','bold');
ylabel('Usage (%)','FontWeight','bold');
```

```
title(sprintf('Paper Usage\nReduction = %.1f%%', paper_red), ...
    'FontWeight','bold','Color',[0 0.2 0.6]);

set(gca,'FontSize',11,'LineWidth',1.2);



% ELECTRICITY

nexttile;
plot(days, electricity, '-o', ...
    'LineWidth', lw, ...
    'MarkerSize', ms, ...
    'MarkerFaceColor',[1 0.5 0]);

grid on;
ylim([70 105]);
xlabel('Days','FontWeight','bold');
ylabel('Usage (%)','FontWeight','bold');

title(sprintf('Electricity Usage\nReduction = %.1f%%', elec_red), ...
    'FontWeight','bold','Color',[0.8 0.4 0]);

set(gca,'FontSize',11,'LineWidth',1.2);


% PLASTIC

nexttile;
plot(days, plastic, '-o', ...
    'LineWidth', lw, ...
    'MarkerSize', ms, ...
    'MarkerFaceColor',[0.2 0.7 0.3]);

grid on;
ylim([70 105]);
xlabel('Days','FontWeight','bold');
ylabel('Usage (%)','FontWeight','bold');

title(sprintf('Plastic Usage\nReduction = %.1f%%', plastic_red), ...
    'FontWeight','bold','Color',[0 0.5 0]);

set(gca,'FontSize',11,'LineWidth',1.2);


% MAIN TITLE

sgtitle('30-Day Sustainability Challenge Dashboard', ...
    'FontSize',14,'FontWeight','bold');
```

The generated dashboard consists of three separate graphs:

1. Paper Usage Graph

This graph illustrates the gradual reduction in paper consumption over 30 days. The trend shows a consistent decrease from 100% to 80%, achieving a total reduction of 20%. This reflects improved paper-saving practices such as reduced printing and increased digital usage.

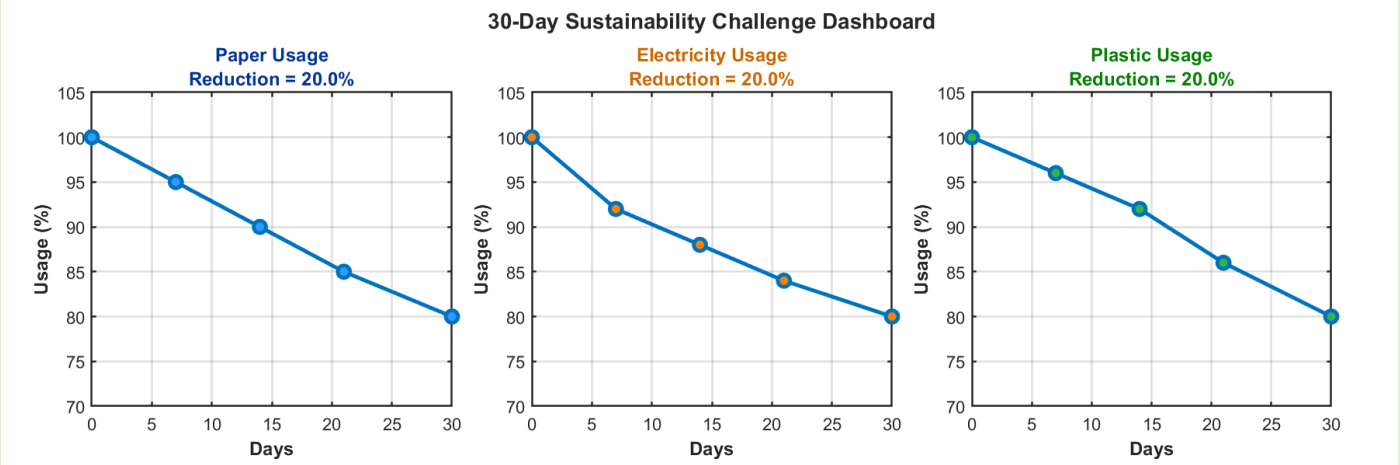
2. Electricity Usage Graph

This graph represents the reduction in electricity consumption over the same period. The usage decreases steadily, indicating the adoption of energy-efficient behaviors such as turning off unused devices and optimizing energy use. The total reduction achieved is 20%.

3. Plastic Usage Graph

This graph shows the decrease in plastic usage across 30 days. The downward trend demonstrates the impact of sustainable habits such as reducing single-use plastics and encouraging reusable alternatives. A total reduction of 20% is achieved.

The simulation successfully demonstrates a 30-day sustainability improvement plan across three key environmental factors. All categories achieved the targeted 20% reduction, indicating that small consistent behavioral changes can lead to significant environmental impact. The dashboard effectively visualizes progress and helps track sustainability performance over time.



- **30-Day Timeline**

- Week 1: Collect data on current paper usage and identify major sources of waste.

- Week 2: Introduce paper-saving practices such as digital documents and double-sided printing.

- Week 3: Monitor implementation and gather feedback from team members.

- Week 4: Compare results with the initial data and evaluate whether the 20% reduction target has been achieved.

- **Expected Outcome**

By the end of 30 days, paper usage should decrease significantly, leading to reduced waste and lower costs.

PRACTICE 2 _(SDGS)

Identify 3 Relevant Sustainable Development Goals (SDGs)

The Sustainable Development Goals established by the United Nations provide a framework for achieving a more sustainable future. Choosing relevant SDGs helps align workplace activities with global sustainability efforts.

After evaluating my sector, I selected three SDGs that are highly relevant.

SDG 12: Responsible Consumption and Production



This goal focuses on reducing waste and using resources efficiently. It supports practices such as minimizing unnecessary consumption and improving waste management.

Achieving SDG 12 is critical because current consumption patterns exceed the planet's capacity, with significant environmental degradation and resource depletion. For example, one-third of all food produced globally is wasted, and by 2050, sustaining current lifestyles could require the resources of nearly three planets. Sustainable consumption and production not only protect the environment but also contribute to poverty alleviation, low-carbon economies, and improved quality of life

SDG 13: Climate Action



This goal emphasizes taking action to combat climate change and reduce environmental impact. In the workplace, this includes reducing emissions, lowering energy consumption, and promoting sustainable behavior.

Achieving SDG 13 requires critical thinking to address the complex challenges posed by climate change. It involves understanding the interconnectedness of climate, health, and development, and recognizing the need for collective action to mitigate the impacts of climate change. By integrating climate strategies into national policies and mobilizing financial resources, countries can ensure that environmental sustainability is a core component of their development efforts. This approach not only protects the environment but also supports economic growth and poverty reduction, making it a critical aspect of sustainable development.

SDG 9: Industry, Innovation, and Infrastructure



This goal encourages innovation and the development of sustainable technologies and systems. It supports implementing smarter and more efficient processes.

SDG 9 is crucial for fostering economic growth, promoting sustainable industrialization, and driving innovation, which are essential for achieving a more equitable and sustainable future.

- **Action Plan**

I will analyze how current workplace practices relate to these three goals. Then, I will identify specific actions that can contribute to each SDG and recommend practical improvements.

- **30-Day Timeline**

- Week 1: Study the selected SDGs and understand their relevance to the workplace.

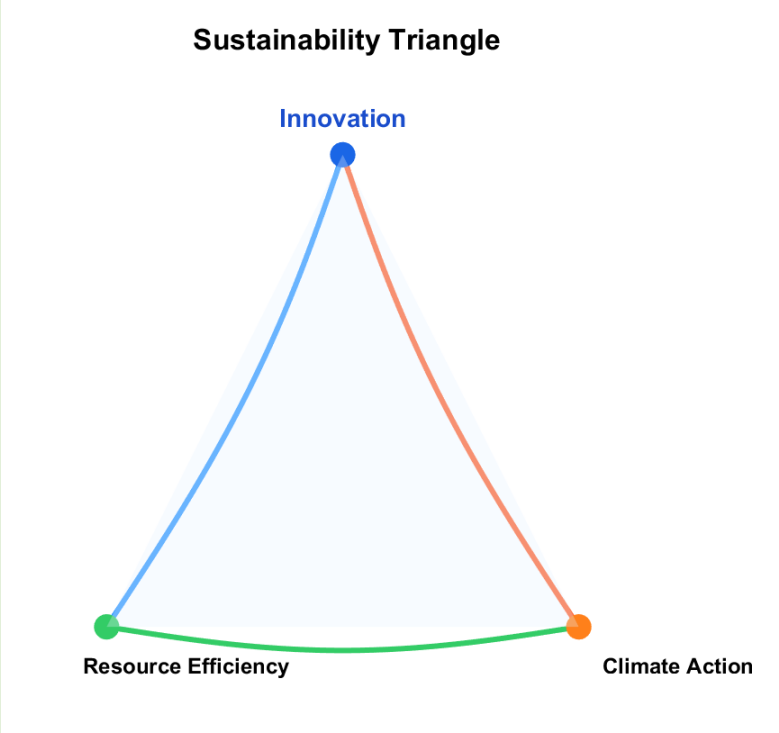
- Week 2: Assess current practices and identify gaps.

- Week 3: Develop recommendations aligned with the SDGs.

- Week 4: Present findings and suggest future improvements.

- **Expected Outcome**

This process will help integrate sustainability principles into daily operations and long-term planning.



PRACTICE3_RECYCLING



Apply a Recycling Policy

Recycling is one of the most effective ways to reduce waste and protect the environment. A proper recycling system can minimize landfill waste and promote responsible disposal of materials.

- **Action Plan**

I will start by identifying the types of waste commonly generated in the workplace, such as paper, plastic, and packaging materials.

Next, I will introduce separate recycling bins for recyclable materials and ensure they are placed in accessible locations. Clear labels and instructions will be provided so employees know how to sort waste correctly.

I will also encourage awareness by explaining the benefits of recycling and how small daily actions contribute to environmental sustainability.

- **30-Day Timeline**

- Week 1: Identify recyclable waste types and prepare recycling stations.

- Week 2: Install recycling bins and share guidelines with employees.

- Week 3: Begin implementation and monitor participation.

- Week 4: Review collected waste and assess effectiveness.

- **Expected Outcome**

The workplace will generate less waste, improve recycling habits, and build a stronger culture of environmental responsibility.

Discussion Questions

1. What is the biggest challenge in implementing sustainability?

The biggest challenge is changing people's habits and increasing awareness. Many employees may resist new systems because they are used to traditional ways of working.

2. What resources do you need?

The most important resources include management support, accurate data, awareness materials, recycling tools, and employee cooperation.

3. How can your team help?

My team can support implementation by following the new practices, reducing unnecessary waste, sharing ideas, and encouraging others to participate actively in sustainability initiatives.